

National University of Computer & Emerging Sciences, Lahore
Department of Electrical Engineering
(FALL 2024)

Notes:

- 1. Submit the hardcopy at the very start of the lecture. Use A4 sheets for the answers.**
- 2. No late submissions. Also submit a scan of the assignment in GCR.**
- 3. Any help from the internet, solution manuals, or any other source is strictly prohibited. Violation may lead to serious consequences.**
- 4. Draw the diagrams where necessary.**

Date of Submission: Monday 24th November 2024

1. Let $D = 4xyax + 2(x^2 + z^2)ay + 4yzaz$ nC/m² and evaluate surface integrals to find the total charge enclosed in the rectangular parallelepiped $0 < x < 2$, $0 < y < 3$, $0 < z < 5$ m.
2. In free space, a volume charge of constant density $\rho_v = \rho_0$ exists within the region $-\infty < x < \infty$, $-\infty < y < \infty$, and $-d/2 < z < d/2$. Find D and E everywhere.
3. Calculate $\nabla \cdot D$ at the point specified if (a) $D = (1/z^2)[10xyz ax + 5x^2z ay + (2z^3 - 5x^2y) az]$ at $P(-2, 3, 5)$; (b) $D = 5z^2 a_\rho + 10\rho z a_z$ at $P(3, -45^\circ, 5)$; (c) $D = 2r \sin\theta \sin\phi a_r + r \cos\theta \sin\phi a_\theta + r \cos\phi a_\phi$ at $P(3, 45^\circ, -45^\circ)$.
4. In a region in free space, electric flux density is found to be $D = \rho_0(z+2d)az$ C/m² ($-2d \leq z \leq 0$) $-\rho_0(z-2d)az$ C/m² ($0 \leq z \leq 2d$) Everywhere else, $D = 0$. (a) Using $\nabla \cdot D = \rho_v$, find the volume charge density as a function of position everywhere. (b) Determine the electric flux that passes through the surface defined by $z=0$, $-a \leq x \leq a$, $-b \leq y \leq b$. (c) Determine the total charge contained within the region $-a \leq x \leq a$, $-b \leq y \leq b$, $-d \leq z \leq d$. (d) Determine the total charge contained within the region $-a \leq x \leq a$, $-b \leq y \leq b$, $0 \leq z \leq 2d$.
5. Let $D = 5.00r^2 a_r$ mC/m² for $r \leq 0.08$ m and $D = 0.205 a_r/r^2$ μ C/m² for $r \geq 0.08$ m. (a) Find ρ_v for $r = 0.06$ m. (b) Find ρ_v for $r = 0.1$ m. (c) What surface charge density could be located at $r = 0.08$ m to cause $D = 0$ for $r > 0.08$ m?

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ELECTROMAGNETIC THEORY

Assignment # 2 on CLO # 02

Name:	Roll Number:
Date of Submission: November 24th, 2024	Section:

Do not write below this line. Do not write below this line.

Instructor: Dr. Huzaifa Rauf

Instructions:

1. Assignment must be handed over to the instructor at the very start of the class.
2. Late submission will not be accepted.
3. Only submit your own work. Do not copy from others. Plagiarism will be dealt with seriously and it will be treated according to the University disciplinary rules.
4. Submit assignment on **A4 size paper**.
5. Use blue or black ink only. Figures may be in color.
6. Handwriting must be legible. Otherwise, your assignment may be returned ungraded.

Problems ↓ [Marks]	Score	CLO # 02				
		Exemplary (5)	Proficient (4)	Developing (3)	Beginning (2)	Novice (1)
P1 [10]						
P2 [10]						
P3 [10]						
P4 [10]						

***Oath:** I have solved all the questions by myself and taken no help from the internet, solution manuals, colleagues, or any other unfair means. It is my work only.*

Signature: _____

Attach this sheet to the front of submitted work.